9 **(a)** **(i)** Explain how X-rays are produced for use in medical diagnosis.

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.....

..... [3]

(ii) State why X-ray images are taken of multiple sections of the body during computed tomography (CT) scanning.

.....

..... [1]

(b) An X-ray image is taken of the structure shown in Fig. 9.1.

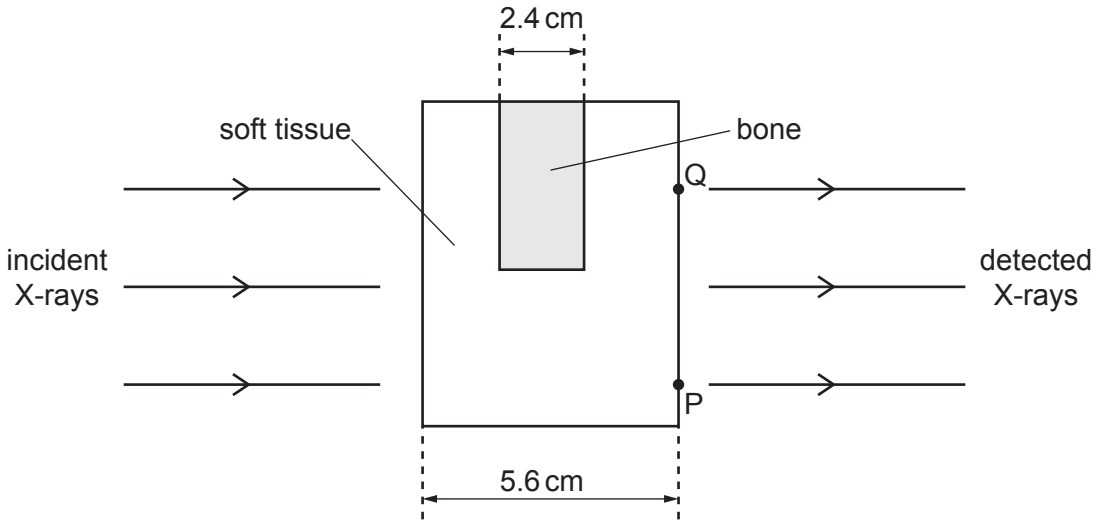


Fig. 9.1

The linear attenuation coefficient of bone is $3.4\,\text{cm}^{-1}$.
The linear attenuation coefficient of soft tissue is $0.89\,\text{cm}^{-1}$.

The incident X-rays are parallel and have a uniform intensity I_0 across the structure.

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Determine, in terms of I_0, the intensity of the detected X-rays from:

(i) point P

detected intensity = ... I_0 [2]

(ii) point Q.

detected intensity = ... I_0 [2]

(c) Explain, with reference to your answers in **(b)**, whether the X-ray image of the structure in Fig. 9.1 has good contrast.

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..... [1]

[Total: 9]